

Process Design Proposal to Synthesize 25,000 tonne/year Dimethylamine Through Modelling Methylamine Synthesis

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1 Abstract

In aiming to create a process design that produced 25,000 tonnes/year Dimethylamine, we formulated a design schematic with a reactor and multiple columns in series in order to separate the products of our reactor. First, an equilibrium reactor was modeled that reacted Ammonia and Methanol to create an equilibrium mixture of Methylamine (MMA), Dimethylamine (DMA), and Trimethylamine (TMA), followed by 3 distillation columns to remove excess ammonia from the reactor outlet. From there the bottoms streams were sent with excess water to a RadFrac extractive distillation column to remove and recycle TMA, the bottoms stream to a flash drum to remove excess water, and a final distillation column to separate the MMA and DMA products. Based on the initial analysis, it has been found that an equal-weight feed of 130,000 tonnes/year each of Ammonia and Methanol produced approximately the target amount of DMA, economic analysis of potential profits included a Gross Margin of 6031 MM\$/oper-yr and an NPV of 9597 MM\$ after an operation of 10 years. In the future, we hope to optimize our recycle streams further, as well as include heat exchangers, particularly in the condenser and reboiler sections of our columns, in order to save on utilities charges

2 Introduction

2.1 Industry/Market Background

The three methylamine products synthesized in our reactor (Methylamine, Dimethylamine, and Trimethylamine, known as MMA, DMA, and TMA, respectively) all have unique uses in the chemical industry; MMA is used in pesticides and explosives, DMA is used in water treatment and in detergents, and TMA is used as an animal food additive, such as in fish food.¹ However, MMA and DMA in particular are used as intermediate chemicals in many large scale chemical processes which increases the industry demand for these particular products as a way of saving time and money.

Current reactor equilibrium technology gives an approximate equilibrium ratio of MMA to DMA to TMA of 5:6:9,² however since the rate of consumption of MMA:DMA:TMA is closer to 1:3:1 in the U.S. and Europe and 1:8:1 in China,³ this means DMA is in higher demand than its co-products, leading to a need for more efficient and cost effective processes of producing DMA while minimizing the other 2 methylamines produced. Currently, size-specific catalysts that minimize the production of TMA⁴ and membranes that remove excess H₂O in the Ammonia/Methanol reactor (which leads to equilibrium ratios favoring more substituted Methylamines)⁵ are two major areas of research as ways of potentially shifting the equilibrium ratios of these three species in the reactors they're synthesized in. By preventing TMA/MMA production, it drastically lowers the downstream costs of extractive distillation and multiple column extractions, however for the purposes of this simulation we're focusing on using current known technology to produce a desired amount of DMA efficiently and in a cost-efficient manner.

2.2 Design Goals and Previous Work

The main goal of this project is to synthesize approximately 25,000 tons/year of dimethylamine and optimize our process to maximize the production of DMA while minimizing the produc-

tion of side products (we chose not to include analysis of other possible side products beyond those shown in Eq. (5)-Eq. (10), such as the reaction of ammonium hydroxide¹ due to their negligible effect on yield) and minimizing the cost of the whole process to meet the global demand for a product (DMA) that is in high demand yet relatively difficult to make due to the azeotropic nature of the equilibrium MMA/DMA/TMA mixture produced by the reactor. This project also aims to produce these products in high purities to meet industry demands.

Previous work (Tab. 20, Fig. 7) was able to create a proposed process design that accomplished the target goal of DMA production both in quantity and purity, however it wasn't economically viable with a low gross margin without including other less visible costs (equipment/installation costs, employment and maintenance charges, etc.). This this updated report is focused on calculating a much clearer economic analysis of this process and beginning to optimize stream results and costs.

3 Design Results

3.1 Basis of Design

Our main goal for the design of this process was producing 25,000 tonnes/year of DMA at 95% purity, and to do so in a cost and energy efficient and profitable manner. For our proposed plant, we chose to locate it in the US Gulf Coast, and while we had very few strict limitations in design, chose to follow and tweak literature's suggestions of column order and roles,¹ and optimized our process with the intent of maximizing potential profit.

3.2 Assumptions and Overall Process Model Parameters

We chose *Amines* as the property method for this simulation after working through the Aspen Property Method Selection Assistant. We assumed the only reactions occurring were those listed in Eq. (5)-Eq. (10), in the creation of MMA, DMA, and TMA. Our process requires 25,000

tons/year of DMA to be produced with the assumption that all equipment is 98% efficient. All other assumptions will be discussed when relevant to specific unit operations (i.e. specific columns or pumps).

3.3 Full Process Flow Diagram

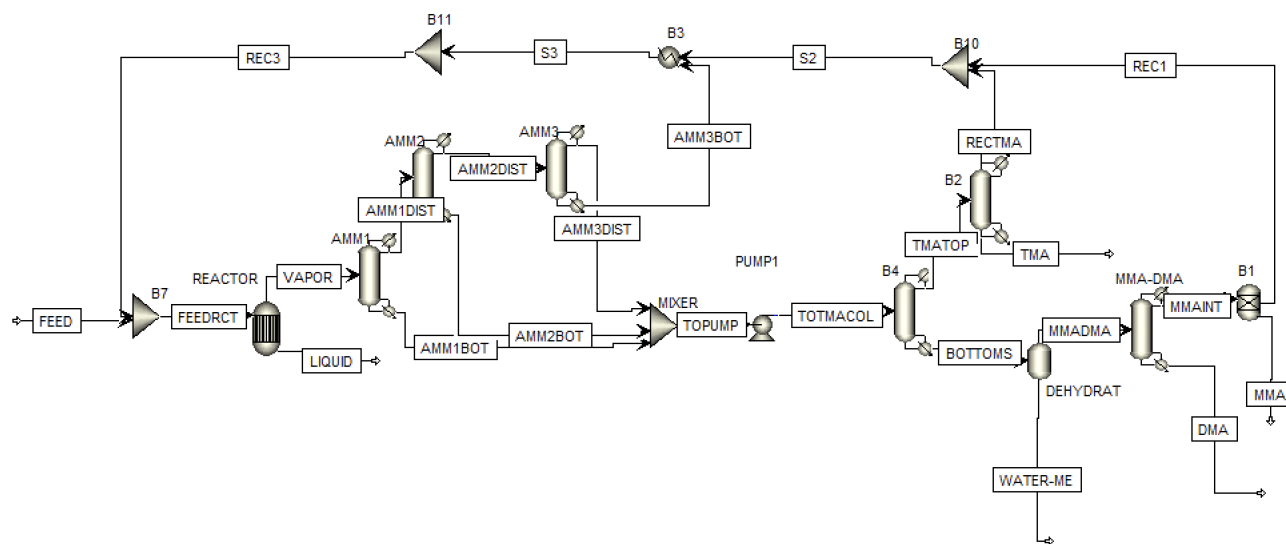


Figure 1: Overall Process Flow Diagram with stream tables
Tab. 10.

3.4 Reactor

An equilibrium reactor will be used in this analysis to determine the required composition with the end goal of producing 25,000 tonnes/year of DMA. Since the software itself doesn't have the database to input the specific catalyst into the reactor, using this type of reactor negates this problem. The other reason is due to the reversible reaction as seen at Eq. (5) - Eq. (10). In terms of the concentration, both forward and reverse rates will be the same for both reactions, thus creating an equilibrium phase at one point.⁶ With the slow, but continuous reaction occurring in this synthesis, an equilibrium reactor provides the best analysis of the system.⁷

3.4.1 Operation Condition

Before going into more detail, there are several assumptions that were made to satisfy the DMA production:

1. Due to the system focusing on the methylamine synthesis to produce three amine groups, the method that will be used in Aspen Plus is assumed to be Amine.
2. Depending on the licensor, a typical chemical plant operates roughly about 333 days or 8,000 hours every year, thus the unit of the components will be per operation/oper-year to accommodate this assumption.⁸
3. A catalyst comprised of carbon materials is supposed to be inserted into the reactor to increase the chemical reaction which affects the energy requirement of the reactor.⁹ In this work, Aspen Plus will model the reactor without the presence of a catalyst.
4. Side reaction is not included during the methylamine synthesis. In reality, this will affect the output of the overall product by a certain amount.¹

All of these assumptions are considered based on the capability of the software and real-life scenarios. Note that due to the assumptions that were made, the simulation results might differ vary from the actual results.

3.4.2 Aspen Plus Simulation Results

With the addition of a recycle stream, the feed stream composition will be based on the previous work in [Tab. 10](#). This will help analyze the effects of adding the recycle stream better. Upon running the simulation in Aspen Plus, the result is found to be:

Upon installing the recycle stream, the production of DMA has unexpectedly massively decreased. MMA still dominates the synthesis with the ratio of MMA:DMA:TMA found to be 84:15:1 respectively. The decrease in DMA production is due to the equipment having all 98% efficiency and some warnings that were displayed during running the simulation – Some

Table 1: Simulation product upon running the reactor under 344 K and 1 atm with $\text{NH}_3/\text{MeOH} = 1$. DMA's target production is not satisfied with the addition of recycle stream, in which MMA dominates the production instead. Note that the inlet reactor is from the feed and recycle streams.

Component	Reactant (tonne/oper-year)	Product (tonne/oper-year)
Ammonia	115,000	42,272.27
Methanol	115,000	601.64
MMA	0	38,553.56
DMA	0	7,439.02
TMA	0	669.62
Water	0	5,521.94
Energy Required		46,645,597.5 Btu/hr

iterations and optimization failed during running the simulation which affects the end result. This resulted in some DMA composition going to undesired streams, thus lowering the DMA production at the final stream.

Component	Reactant (tonne/oper-year)
MMA	62,685.03
DMA	18,010.89
TMA	1,085.06

Table 2: The majority of the three-amine groups failed to be filtered out by the other column.

In [Tab. 2](#), there's still considered a large amount of MMA, DMA, and TMA that goes back into the inlet stream of the reactor. Combined with the final stream value in [Tab. 1](#), DMA will almost reach the required target production. With the majority goes back into the recycle stream, further optimization is required.

A sensitivity analysis through Aspen Plus is then initiated to fix the problem by finding the optimal ratio of $\hat{\text{NH}}_3/\text{MeOH}$.

It is founded that the optimal ratio of NH_3/MeOH is 0.25 upon using the sensitivity tools in Aspen Plus. Note that the 25,000 tonne/oper-year target production hasn't been reached.

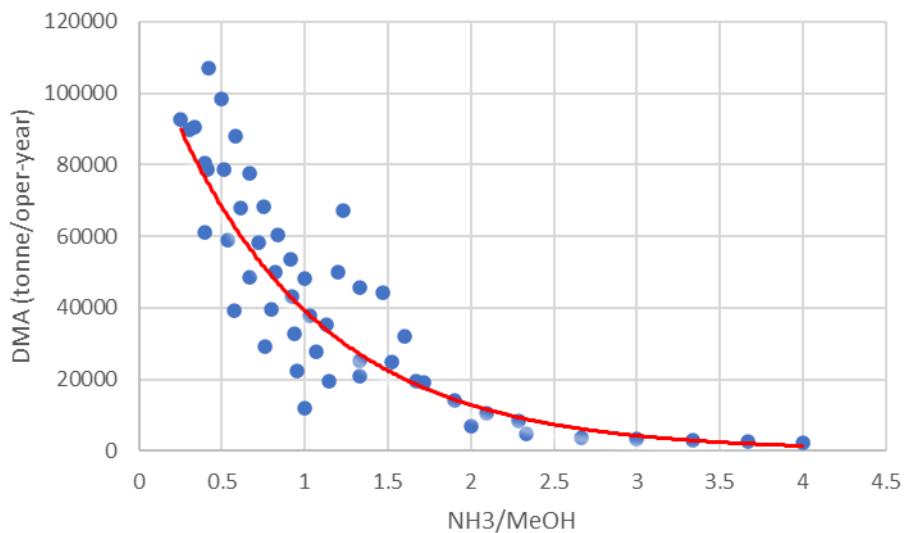


Figure 2: DMA production has an exponential decrease trend as NH_3/MeOH ratio increase.

Using some trial and error, the necessary feed composition in [Tab. 17](#) is determined.

Further optimization of pressure and temperature was done using the sensitivity tools in Aspen Plus. Upon [Fig. 8](#) and [Fig. 9](#) analysis, a lower temperature and pressure are considered to get the maximum production of DMA. Thus, 400 K and 1 atm operation conditions are used.

Table 3: Optimal product output upon using sensitivity analysis from Aspen Plus.

Component	Reactant (tonne/oper-year)	Product (tonne/oper-year)
Ammonia	58,500	8,211.03
Methanol	146,250	2,835.29
MMA	0	33,459.67
DMA	0	25,617.88
TMA	0	9,245.14
Water	0	3,229.12
Energy Required		11,030,629.3 Btu/hr

With these optimizations, some values can be lowered which increases the profitability of this proposed system. Looking at the energy consumption of the reactor, approximately

30,000,000 Btu/hr can be saved. As a plant typically operates for 5-10 years, this will hugely reduce the cost of energy consumption. The required NH_3 is also reduced by a factor of 2, however, this optimization increases the required amount of MeOH by 30,000 tonne/oper-year. This is still considered fine as the price of MeOH is significantly lower than NH_3 – More detailed analysis will be discussed in a further section.

This optimization, however, doesn't fix the large excess of the three-amine group that goes into the recycle stream. [Tab. 19](#) shows that the majority of DMA still goes back to the recycle stream instead to the desired stream. However, this optimization is able to reduce the ratio of the MMA, DMA, and TMA that goes back to the stream from 62:71:62 to 52:58:44 respectively. Even so, further optimization in the future will be very beneficial to maximize the profitability of this proposed system.

3.5 Ammonia Columns

3.5.1 Operation Condition

The first column in the train of separation after the reactor is the ammonia column shown in [Fig. 4](#). This is done by using 3 distillation columns to separate ammonia (NH_3) from the other components by Aspen Plus. The distillation columns are DSTWU with the settings in [Tab. 4](#), where Amm1 is the first column with partial condenser and all vapor distillate chosen from¹⁰ whereas the other two columns use total condensers since we would do not want vapor distillate for the other separations.

3.5.2 Aspen Plus Simulation Results

Initially, the goal of 98% removal of ammonia was difficult to find based on suggested operating column pressures.¹ Using Aspen Plus, higher pressures were found to be needed in order to find the bubble point temperature. Once this did not have errors, the streams showed that this only separated methanol and water into the bottoms stream from the other components. Since

this project's goal is to extract DMA and the byproducts MMA and TMA, further separation is needed. The second column separates MMA and DMA into the bottoms stream. Once these were separated, it was found through Aspen Simulation, that TMA was in the distillate stream along with NH_3 , which was expected. Since the boiling point of TMA is higher than both MMA and NH_3 , it would be expected that it is less volatile, therefore moving towards the bottom stream. The third distillation column separates NH_3 into the bottoms and TMA into distillate streams where TMA was used as the light key as it was still found to be more volatile despite being isolated with NH_3 , a molecule with a much lower boiling point. The NH_3 bottoms stream is sent to be removed from this process, while the other streams are mixed to move on in this distillation train. The column pressures were based on literature.¹⁰ Tab. 5 summaries this ammonia extraction process over the three distillation columns.

While using more distillation columns may lead to higher energy usage operating costs, this proposed solution has provided 97.8% removal of ammonia before adding the recycle stream. With the recycle stream, only 60% of ammonia is removed, with less than 2% or negligible loss of MMA, DMA, and TMA, being crucial to reach the project goal while maintaining purity.

3.5.3 Critical Parameters Summary Table

Table 4: Ammonia distillation columns settings, pressure in psia.

Setting	Ammdist1	Ammdist2	Ammdist3
Heat Duty (Btu/hr)	3.0e7	3.9e7	4.6e7
Feed Stage	1	66	3
Actual Stages	7	70	3
Minimum Reflux Ratio	0.728	2.14	152
Condenser Pressure	450	250	250
Reboiler Pressure	455	255	255

Table 5: Inlet and outlet flow rates of the inlet stream, distillation outlet (recycled ammonia), and the bottom outlet (mixer of the three columns) for ammonia columns

Component	Inlet (tonne/oper-yr)	Distillate (Mixer) (tonne/oper-yr)	Bottoms (tonne/oper-yr)
Ammonia	55823	33490	22332
Methanol	18237	0	18237
Water	129441	0	129441
MMA	89497	0	89497
DMA	68317	0	68317
TMA	28666	3960	24706

3.6 TMA Extractive Distillation Column

The TMA column (with critical parameters in [Tab. 8](#) and flowrates in [Tab. 22](#)) is a Radfrac Column due to the azeotropic nature of the MMA/DMA/TMA mixture. From literature, additional H₂O is usually added before being fed into the column to help separate the boiling points of the species, however through Sensitivity analysis shown in [Fig. 5\(b\)](#), additional H₂O was actually detrimental to the process, and made optimizing the distillate rate and adjusting it to an acceptably low amount incredibly difficult. With the deletion of an additional H₂O stream, we were able to reduce the Distillate Rate from 550 lbmol/hr to 65. The inclusion of the Bottoms temperature is simply for practical reasons, as the column tended to error when the outlet temperature dipped below 0°F, and including it proved a useful visual tool. Literature advised a 50 stage column,¹¹ with operation between 2-3 MPa, and while we opted for a 3 MPa column pressure, through troubleshooting we found that a column with much fewer stages was actually optimal in preventing stage drying and saving costs. High Distillate Rates and Reflux Ratios were also necessary to prevent drying, and were chosen based on sensitivity analysis found in [Fig. 5\(a, c\)](#)

3.6.1 Critical Parameters/Stream Results Summary Table

Table 6: Inlet and outlet flow rates to the TMA extraction column (Including additional water to resolve the issues with the azeotropic nature of the 3 methylamines mixture)

Component	Inlet (tonne/oper-yr)	Bottoms (tonne/oper-yr)	Distillate (tonne/oper-yr)
Ammonia	1530.32	1315.54	214.785
Methanol	4813.2	4809.06	4.14026
Water	61951.1	61951.1	7.9069e-06
MMA	60634.3	57980.8	2653.53
DMA	25679	22984.7	2694.36
TMA	5941.25	0.0296195	5941.22

Table 7: TMA extraction column critical parameters summary table

Number of Stages	8	
Feed Stage (on)	6	
Pump Discharge Pressure	45	bar
Reboiler type	Kettle	
Reboiler duty	3.345e+6	Btu/hr
Condensor type	Total	
Condensor duty	-7.401e+6	Btu/hr
Operating Pressure	3	MPa
Distillate Rate	65	lbmol/hr
Reflux Ratio	15	(molar)

(a) First subtable

Stream	Temperature	Pressure	Phase	Total Mass Flowrate (tonne/oper-yr)
Inlet	182.25°F	652.67 psia	Liquid	160549
Distillate	212.798°F	435.113 psia	Liquid	11508
Bottoms	53.41°F	435.113 psia	Liquid	149041

(b) Second subtable

3.6.2 Notes on changes from Recycling

The addition of a recycling stream proved to be rather complicated for this process, and due to this the temperatures reported here are for the process without the recycling stream, as further optimization is needed to adjust the inlet temperature of the TMA column with recycled materials. The main difference is that the inlet temperature is approximately 100°C lower than with the recycling stream as seen in our finished stream results in [Tab. 10](#), and because of that Aspen has a much harder time properly executing the extractive distillation column, thus the

negative temperatures and errors seen when our process is ran. Due to time constraints we see this as room to improve, as recycle streams are incredibly important in cost and energy saving measures for industrial plants, especially in this process where TMA can be dismutated into DMA - our target product.

3.7 Dehydration Column and MMA/DMA Seperation Column

The dehydration column and MMA/DMA columns are designed to remove the residual water, methanol, and separate the MMA from DMA.¹ For the first dehydration column, the flash column is selected because of the significant difference between the boiling point temperatures of methanol (64°C)¹²/water (100°C) and MMA (-6.5 °C)³/DMA (7 °C).¹³ The flash column has a simpler structure compared to the conventional distillation column and therefore it shows an advantage in design and the overall cost.

The dehydrated MMA/DMA stream is then sent to the distillation column to further separate the MMA from DMA.¹ A DSTWU distillation column is chosen for this process in consideration of the similar physical properties of DMA and MMA. The operating pressure of this distillation column is set to 7 bar based on the literature.¹ The recovery rate for the light key (MMA) is set to 0.996 and 0.01 is set for the heavy key (DMA).

3.7.1 Critical Parameters Summary Table

Table 8: MMA/DMA column critical parameters summary table

Number of Stages	651	
Feed Stage (on)	339	
Light Key	MMA	
Recovery	0.996	
Heavy Key	DMA	
Recovery	0.01	
Condenser Pressure	0.7	MPa
Reboiler Pressure	0.7	MPa
Distillate temperaure	46.104	C
Bottom temperature	50.716	C

Table 9: Inlet and outlet flow rates of the inlet stream, waste-water outlet, MMA outlet and DMA outlet

Component	Dehydration Inlet (tonne/oper-yr)	Water Outlet (tonne/oper-yr)	MMA Outlet (tonne/oper-yr)	DMA Outlet (tonne/oper-yr)
Ammonia	1315.54	27.4633	1288.07	5.072e-8
Methanol	4809.06	2996.12	0	1812.94
Water	61951.1	58477.1	0	3474
MMA	57980.8	190.431	57559.2	231.161
DMA	22984.7	1.06536	229.836	22753.8
TMA	0.02962	1.36e-10	0.0296	1.17e-12

3.8 Summary Stream Tables

Table 10: Previous Process Flow Diagram Stream Tables and Operating Conditions. Temperature in °C, pressure in psia.

Component	Feed	REC3	FeedRCT	Vapor
(tonne/oper-year)				
Ammonia	58 500	20 959.2	79 459.2	30 031.6
Methanol	146 250	3130.03	149 380	13 864.7
DMA	0	28 124.7	28 124.7	56 569.1
MMA	0	$1.136\,25 \cdot 10^{-6}$	$1.136 \cdot 10^{-6}$	59 733.7
TMA	0	8875.28	8875.28	29 448.3
Water	0	3127.11	3127.11	79 318.8
Total	204 750	64 216.3	268 966	268 966
Temperature °(F)	260	−25.5725	147.486	260.33
Pressure (psia)	14.6959	14.6959	14.696	14.696

(a) First subtable

Component	Amm1Dist	Amm1Bot	Amm2Dist	Amm2Bot	Amm3Dist
(tonne/oper-year)					
Ammonia	30 028.6	3.003 16	15 014.3	15 014.3	3002.86
Methanol	6321.9	7542.78	3160.95	3160.95	3160.95
DMA	56 569	0.086 87	28 284.5	28 284.5	28 284.5
MMA	59 721.7	11.947	29 860.9	29 860.9	29 860.9
TMA	29 448.3	$4.5075 \cdot 10^{-9}$	14 724.2	14 724.2	11 779.3
Water	0	79 318.8	0	0	0
Total	182 090	86 876.6	91 044.8	91 044.8	76 088.5
Temperature °(F)	209.684	436.8	195.637	195.637	170.95
Pressure (psia)	450	455	367.399	367.399	250

(b) Second subtable

Component	Amm3Bot	ToPump	ToTMACol	TMATop
(tonne/oper-year)				
Ammonia	12 011.5	18 020.2	18 020.2	$9.9956 \cdot 10^{-11}$
Methanol	$1.1569 \cdot 10^{-7}$	13 864.7	13 864.7	$2.1593 \cdot 10^{-19}$
DMA	$1.0352 \cdot 10^{-6}$	56 569.1	56 569.1	$3.606 \cdot 10^{-6}$
MMA	$1.092\,94 \cdot 10^{-6}$	59 733.7	59 733.7	$1.0047 \cdot 10^{-9}$
TMA	2944.82	26 503.5	26 503.5	15 277.4
Water	0	79 318.8	79 318.8	$5.987 \cdot 10^{-21}$
Total	14 956.3	254 010	254 010	15 277.4
Temperature °(F)	98.107	360.4	346.5	285.45
Pressure (psia)	255	250	652.67	435.113

(c) third subtable

Component	TMA	RECTMA	S2
(tonne/oper-year)			
Ammonia	$1.999 \cdot 10^{-11}$	$799\,652 \cdot 10^{-11}$	4507.24
Methanol	0	0	1560.98
DMA	$7.212\,45 \cdot 10^{-7}$	$2.885 \cdot 10^{-6}$	14 141.8
MMA	$2.01 \cdot 10^{-10}$	$8.0375 \cdot 10^{-10}$	$8.0375 \cdot 10^{-10}$
TMA	14 971.9	305.55	3184.85
Water	0	0	1477.87
Total	14 971.9	305.549	24 872.7
Temperature °(F)	285.45	285.45	30.547
Pressure (psia)	435.113	435.113	30

(d) Fourth subtable

Component	MMADMA	DMA	MMAINT	MMA	REC1
(tonne/oper-year)					
Ammonia	17 745.4	13 238.2	4507.24	0	4507.24
Methanol	6310.42	4749.44	1560.98	0	1560.98
DMA	56 567	442 425.3	14 141.8	0	14 141.8
MMA	59 585.3	44 683	14 902.3	0	0
TMA	11 226.1	8346.77	2879.3	0	0
Water	6181.3	4703.44	1477.87	0	1477.87
Total	157 616	118 146	39 469.4	14 902.3	24 567.1
Temperature °(F)	104	65.7454	29.2187	29.2187	29.2187
Pressure (psia)	14.696	60	30	30	30

(e) third subtable

Component	Liquid	Bottom	Water-ME
(tonne/oper-year)			
Ammonia	0	18 020.2	274.757
Methanol	0	13 864.7	7554.26
DMA	0	56 569.1	2.0911
MMA	0	59 733.7	148.398
TMA	0	11 226.1	$4.4838 \cdot 10^{-5}$
Water	0	79 318.8	73 137.5
Total	0	238 733	81 117
Temperature °(F)		-323.634	104
Pressure (psia)		435.113	14.696

(f) third subtable

3.9 Energy Requirements Table

Table 11: Energy Requirements per major unit operation, in Btu/hr

Unit Operation	Energy Requirement (Btu/hr)
Reactor	1.10E+07
Ammonia 1	3.0E+07
Ammonia 2	3.9E+07
Ammonia 3	4.6E+07
TMA (B3) Condenser	5.65E+06
TMA (B3) Reboiler	2.44E+07
TMA Separator	7,060.35
Dehydration Flash	2.68E+07
MMA/DMA	1.65E+06
MMA/DMA Separator	1.52E+06
Total	1.87E+08

4 Discussion

Based on the final products of DMA stream, the goal of 25,000 tonne/oper-yr is fairly reached with the actual DMA production of 22753.8 tonne/oper-yr. The main reason for the remaining difference between the target production and the actual production is the difficulty of separating the TMA from the MMA and DMA as they have close boiling points. This can be resolved by using different distillation columns like the DSTWU column to better separate the mixtures without losing too much of the target products. The other issue that needs to be addressed in future work is the low purity of the final products which is mainly caused by the dehydration column. There is a small amount of water existing in the DMA outlet that lowered the overall purity, and this can be prevented by replacing the flash column with the distillation column. By making the switch, the water will be completely removed from the streamline; however, the overall cost will be inevitably raised as the trade-off.

The energy requirement of this system is another factor that needs to be considered in

future work. The current main source of energy consumption is the reactor which solely uses 8,039,106.05 Btu/hr. This number can be further reduced by changing the reactor settings, and other energy consumption like the pumps and heaters can also be lowered by designing the system more effectively and adding heat exchangers to recycle the heat generated somewhere else. Overall this is a promising plant with room for improvement, and by further optimizing the processes and recycling the side products, it will be economically viable.

5 Sizing and Materials

5.1 Sizing

First, the optimized reactor is then sent to the APEA in Aspen Plus to determine the size of the reactor based on the required parameters.

Table 12: Required sizing parameters based on the optimized operating condition for each piece of equipment. All data were taken from the APEA tool in Aspen Plus.

Parameter	Reactor	Ammonia 1	Ammonia 2	Ammonia 3
Diameter	7.5	9	5	8
Height	30	40	20	32
Tray Spacing		2	2	2

Parameter	Dehydration Flash	TMA	TMA Separator	MMA/DMA	MMA/DMA Separator
Diameter	4	10	3.5	6	3
Height	12	22	12	14	12
Tray Spacing		2		2	

The thickness is then also found by using the equation

$$t = \frac{P_i D_i}{2SE - 1.2P_i} \quad (1)$$

$$t = \frac{P_i D_i}{4SE + 0.8P_i} \quad (2)$$

Where E is the weld efficiency, P_i is the pressure, D_i is the inner diameter, and S is the maximum stress. The larger t will be used as the thickness of the equipment. Assume that $E = 0.98$, the thickness of all equipment is found to be

Table 13: Thickness calculated using [Eq. \(1\)](#) and [Eq. \(2\)](#)

Equipment	t (inch)	Equipment	t (inch)
Reactor	1.18	TMA	0.47
Ammonia 1	1.14	TMA Separator	1.41
Ammonia 2	2.45	MMA/DMA	0.95
Ammonia 3	3.19	MMA/DMA Separator	0.97
Dehydration Flash	1.61		

Note that all of these calculations are based on theoretical equations found in [Eq. \(1\)](#) and [Eq. \(2\)](#) due to Aspen Plus failing to give the thickness of all of this equipment. The results are based on all combined optimization done from all equipment. Since the equipment is dependent on each other, changing one parameter could result in changing the size and thickness of all equipment.

5.2 Material Choices

Methylamine synthesis could produce a corrosive species during the reaction. During the reaction, some methyl species become radicals which react highly with strong oxidants, thus making them very corrosive.¹⁴ Stainless steel possesses a very strong resistance to corrosion, making the material a very suitable candidate to handle methylamine synthesis.¹⁵ As a result, 304 stainless steel will be the proposed material to be used in all equipment

The material for the pipe needs to have a high resistance to corrosion as well. As a result, stainless steel pipe would be an ideal candidate to be used in this proposed system as well. However, two other materials, PTFE and PVDF, are also strong candidates to be chosen as well. Both are universally used in industry due to their strength, durability, and resistance to corrosion, making them perfect for transporting high-strength acids and chemicals.¹⁶

6 Economic Analysis

Tab. 18 will be used as a guideline to determine the gross margin of this proposal.

Table 14: Gross margin is calculated by subtracting the total cost of the raw material by the total revenue, $\text{Gross margin} = \text{total raw material} - \text{total revenue}$.

Component	Value (tonne/oper-year)	Price (\$/oper-year)
Ammonia	58,500	24,255,855,000
Methanol	146250	8,678,796,750
Total Raw Material Cost		32,934,651,750
Ammonia	33,490	13,885,958,700
MMA	11,159	597,921,538
DMA	25,617	9,990,630,000
TMA	12,105	12,105,000,000
Total Revenue		38,966,159,336
GROSS MARGIN		6,031,507,586

The gross margin shows a positive value, showing that the proposed system can generate a profitable income each year. Note that Tab. 14 doesn't consider any external factors such as

energy, tax, and employees. These factors have a high impact on the net profit of this proposed system. Some of these factors will be accounted for in the next section. To maximize the accuracy of this proposed system, future analysis that compensates for all of these factors needs to be done.

6.1 Fixed Capital Costs

Table 15: Fixed equipment cost calculated using Aspen

Equipment	Installed Cost (\$)	Equipment	Installed Cost (\$)
Reactor	48,600	TMA	65,900
Ammonia 1	100,200	TMA Separator	26,400
Ammonia 2	100,200	DMA Separator	104,300
Ammonia 3	100,200	MMA Separator	17,100
Dehydration Flash	36,400		

Table 16: Fixed installed cost calculated using Aspen

Equipment	Installed Cost (\$)	Equipment	Installed Cost (\$)
Reactor	191,200	TMA	227,800
Ammonia 1	284,900	TMA Separator	115,800
Ammonia 2	284,900	DMA Separator	287,200
Ammonia 3	284,900	MMA Separator	123,000
Dehydration Flash	127,800		

The fixed capital cost is calculated in Aspen and the installed capital is shown in [Tab. 16](#), with the equipment cost shown in [Tab. 15](#). For the Ammonia 2 and Ammonia 3 distillation columns, the installed costs are estimated based on the Ammonia 1 column given the similar

sizes they have. The enclosed agitated tank is used as the main reactor, while all the ammonia columns use multiple-diameter packed towers. For the first TMA distillation column and DMA separator, multiple trayed columns are assumed, and for MMA separator and TMA separator, the vertical process vessels are used. The overall fixed cost is \$1,927,500.

6.2 NPV Calculations

The Net Present Value (NPV) of the project

$$NPV = \sum_{n=1}^{n=t} \frac{CF_n}{(1+i)^n} \quad (3)$$

where CF_n is the cash flow in operating year n , t is the project life in years, and i is the interest rate or cost of capital in percentage. The cost of capital can be calculated using

$$i_c = (DR * i_d) + ((1 - DR) * i_e) \quad (4)$$

where i_c is the cost of capital, DR is the debt ratio, i_d is the interest rate due on debt, i_e is the cost of equity. Since this is still a proposal, we shall assume there is no cost of equity and the cost of capital or interest rate is 12%.

This plant operation assumes a MARCS depreciation with a five-year recovery period with corporate income tax at 21%. The plant assumes a 10 year period with the first two in fixed capital expenses split in half, and installment expenses split 75% and 25%. Operation starts after year 1. Total working capital is paid back at the end of operation. Installment expenses are from APEA analysis of the column units and overall fixed cost is calculated from [Tab. 16](#) as \$1,927,500.2

The product revenues are calculated in [Tab. 14](#).

The operating and utility costs are determined by APEA where the energy consumption in electricity, Btu/hr is calculated for MM\$/year. The operating cost comes out to 973 MM\$/year

and utility cost of 2.735 MM\$/year.

This yields a NPV value after operation of 10 years, in year 11 of 9597 MM\$ shown in [Fig. 6](#)

6.3 Discussion of Economic Analysis

This project is overall profitable over a 10-year period with 2 years in building the process. It has a positive NPV value which tells us the difference between present values of cash inflow right now and cash outflows over time. This is also assuming a 12% interest rate or cost of capital. These values may change in operating costs and installment or material costs largely based on the location of this process. The operating costs could be improved in which the distillation train could have less rigorous separation for lower cost since a negligible amount below 2% can already be separated using less electricity. Although this economic analysis is unable to include fluctuations in profit from the wages of employees and possible investors, the gross margin is very large which makes this proposal effective as it reaches the goal of manufacturing 25,000 tonne/operating year while being profitable despite the material costs.

7 Safety and Environment

7.1 P&ID for the Entire Process

This process requires safety controls around the main processes which would be the reactor and the furnace. The reactor may produce a highly exothermic reaction which would need to be controlled, and the furnace may leak harmful toxins to the air or activate flammable materials. In this proposal, the safety additions include a pressure switch, two level valves paired with a low and high level alarm shown in [Fig. 3](#) around the furnace B3 and the reactor REACTOR.

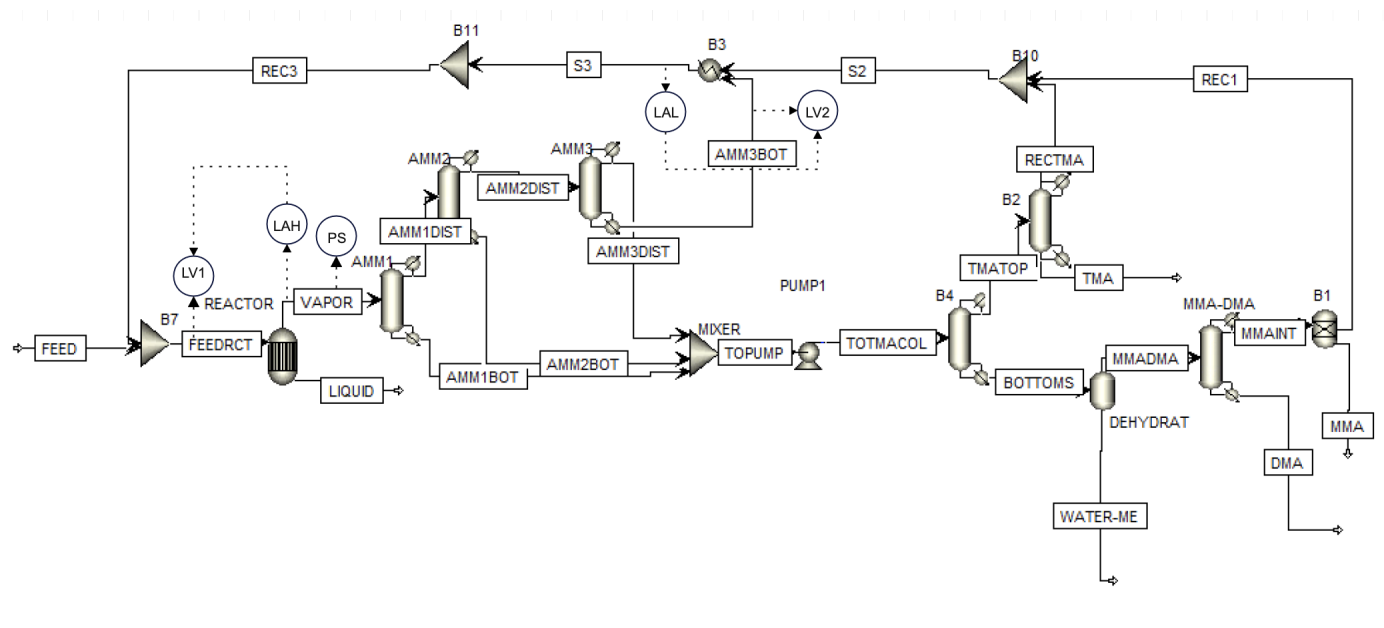


Figure 3: P&ID with safety controls on major equipment.

7.2 Environmental Considerations

The Dimethylamine synthesis process will generate wastewater that contains ammonia and methylamines as the major pollution to the environment. The recycle streams have been introduced to the system to maximize the overall production efficiency and also minimize liquid pollution. However, due to the great mass of the entire production process, there is still over 5,521.94 tonne/oper-yr of wastewater mixed with methanol, ammonia, and other residual methylamines. This waste liquid needs to be properly treated especially because ammonia is commonly recognized as one of the toxic chemicals. Given this large amount of liquid waste, extra recycling might be the optimal way that reduces the impact on the environment. By constructing extra separators or using reverse osmosis, the purified ammonia and water may be obtained from the process and can be reused as fresh feed. On the other hand, If recycling is not economically viable, the last solution is to direct dispose of the wastewater after diluting it to the concentration that is permitted by the law.

7.3 Global/Societal Impacts

Like what has been discussed in the introduction section, the main global impact of this Dimethylamine synthesis is the significance of DMA as an organic solvent and additive in water treatment. The high demand for DMA in the U.S., Europe, and China made the local supply source of DMA especially critical. By constructing local chemical plants for DMA production, the high transportation cost can be avoided as well as the safety risk given its high flammability. The byproducts of this plant can also be sold as the raw material for pesticides and animal food additives. Therefore, this plant is preferred to be located near other secondary process factories or downstream chemical plants. Also, considering the high energy consumption and global warming crisis we are facing right now, reducing the overall energy requirement will be one of the most important goals for the future optimization of the process. The heat exchange system will have the greatest potential for further improvement.

8 Conclusions/Recommendations

This project was incredibly insightful into the difficulties and intricacies of designing industrial scale separations processes, and just how many working parts go into even the smallest detail; the smallest section of pipe connecting two distillation columns can have a team of people poring over what material, thickness, coatings, length, support structure, manufacturer, and whatnot the pipe should be made of. By using guidance from literature and formulating a process that uses multiple ammonium extraction columns, a TMA extraction column and recycling step, and various unit operations to separate and purify our DMA and MMA products, we were able to make a process that made and purified 25,000 tonne/oper-year of DMA, as per our project guidelines. The annual operating cost of our final design was 973 MM\$/yr with an NPV value after 10 years of 9597 MM\$. The gross margin was also found to be 6031 MM\$/oper-yr. Adding a recycle stream certainly helped in bringing down the raw materials cost of this project, as without the recycle stream it came out to 32934 mm\$/oper-yr. Future ar-

tasks of work in this project includes implementing heat exchangers, particularly before certain unit operations such as the reactor and the TMA column, and further optimizing the recycle stream, as in its current state it increases the temperature of several unit operations (such as the TMA column and a couple of the ammonium extraction columns) that cause errors and non-ideal results.

9 Project Plan

Task	Planned Start Date	Planned Finish Date	Responsible Person	Completion	
Project Plan/Gantt Chart	4/6/22	4/7/22	All		
Interim Project Report	4/17/22	5/5/23	All		
Final Project Report	5/8/23	6/9/23	All		
Project Poster	5/13/23	6/13/23	All		
Project Poster Presentation	5/13/23	6/14/23	All		
Research					
Finalizing Aspen Results between sections	4/17/23	5/5/23	All		before wk6
PFD Diagram	4/17/23	4/24/23	Brian		
Aspen Results Energy Analysis	4/24/23	5/1/23	Anna		
Aspen Results Stream Table	4/24/23	5/1/23	Anna		
Optimization	5/6/23	5/12/23	Jovan		after wk6
Equipment Sizing			Jovan		
Equipment Materials	5/21/23	5/28/23	Jovan		after wk3
Economic Evaluation			Kunpeg Yu		
NPV Evaluation	5/21/23	5/28/23	Brian		
HAZOP Analysis	5/14/23	5/21/23	Anna		after wk2
P&ID for Controls	5/12/23	5/21/23	Brian		
Environmental Analysis	5/19/23	5/26/23	Kunpeng Yu		after wk8

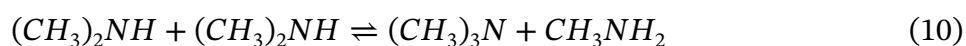
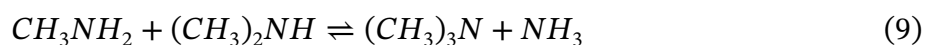
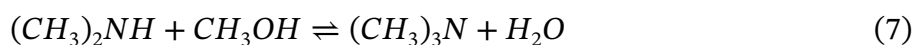
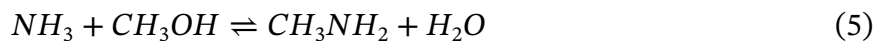
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10 Appendix

10.1 Methylamines Synthesis Equations



10.2 Ammonia Separation Process

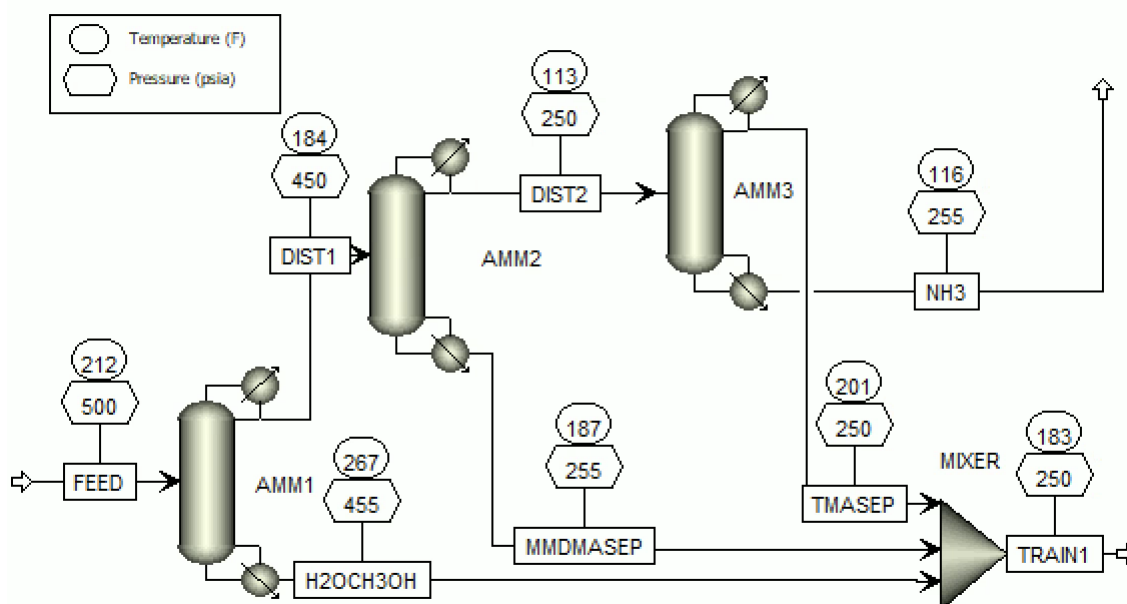


Figure 4: Train of distillation columns to separate ammonia from the other products.

10.3 Sensitivity Analysis for TMA Extractive Distillation



Figure 5: Sensitivity Analysis for TMA Column, altering Distillate Rate, Added H₂O, and molar Reflux Ratio against Distillate TMA, Bottoms DMA, and Bottoms Temperature (the last variable is simply to find the limit where the column no longer errors)

10.4 Optimized Feed Composition

Table 17: Required feed composition to achieve the DMA target production with $\text{NH}_3/\text{MeOH} = 0.25$.

Component	Flow Rate (tonne/oper-year)
Ammonia	58,500
Methanol	146,250
Water	0
MMA	0
DMA	0
TMA	0

10.5 Material Cost

Table 18: The cost of all material is taken from the website, Sigma Aldrich.¹⁷

Component	Pack Size	Price (\$)
Ammonia	800 mL	271
Methanol	4 L	188
MMA	1 L	36.5
DMA	1 Kg	390
TMA	100 g	184
Water	4 L	137

Component	Reactant
	(tonne/oper-year)
MMA	35,798.13
DMA	35,866.57
TMA	7,316.01

Table 19: Most of the three-amine group still failed to be filtered by the system. However, the ratio of the amine groups that goes to the recycle stream is lower than in prior optimization.

10.6 Net Present Value

		Intallment Expenses		1 MACRS		21% tax rate based on prev yr income		CF after tax = P(1-t)+Dt					
		Fixed Capital Cost		1.927 5 years recovery		tax rate		0.21		P-grossprofit = main			
		Life		10 years		(MM\$) Million		interest rate		0.12		product revenues -ccop	
Year	Installment Expenses	Total Fixed Capital Expenses	Gross Margin (MM\$)	Operating and Utilities Cost	Gross Profit (MM\$)	Depreciation Schedule (% of fixed capital)	Depreciation Charge (MM\$)	Taxable Income (MM\$)	Taxes Paid (MM\$)	Cash Flow (MM\$)	PV	DF	NPV
0	-0.75	-0.96								-1.71	-1.71	1.00	43.70
1	-0.25	-0.96								-1.21	-1.10	0.91	43.70
2			6031	975.735	5055.27	20.00	1.06	5054.21	0.00	5055.27	4177.90	0.83	43.70
3			6031	975.735	5055.27	32.00	1.69	5053.58	1061.38	3993.88	3000.66	0.75	1105.08
4			6031	975.735	5055.27	19.20	1.01	5054.25	1061.25	3994.01	2727.97	0.68	2166.33
5			6031	975.735	5055.27	11.52	0.61	5054.66	1061.39	3993.87	2479.88	0.62	3227.72
6			6031	975.735	5055.27	11.52	0.61	5054.66	1061.48	3993.79	2254.39	0.56	4289.20
7			6031	975.735	5055.27	5.76	0.30	5054.96	1061.48	3993.79	2049.44	0.51	5350.68
8			6031	975.735	5055.27	0.00	0.00	5055.27	1061.54	3993.72	1863.10	0.47	6412.22
9			6031	975.735	5055.27	0.00	0.00	5055.27	1061.61	3993.66	1693.70	0.42	7473.83
10			6031	975.735	5055.27	0.00	0.00	5055.27	1061.61	3993.66	1539.73	0.39	8535.43
11		1.93	6031	975.735	5055.27	0.00	0.00	5055.27	1061.61	3995.59	1400.43	0.35	9597.04
Total					50552.65	100		50547.37	9553.3421	40998.308	23184.39	1.00	

Figure 6: Spreadsheet of calculations for NPV over estimated plant life including estimated total capital investment, annual raw material operating costs, annual product revenues, and annual utilities operating costs.

10.7 Previous Work

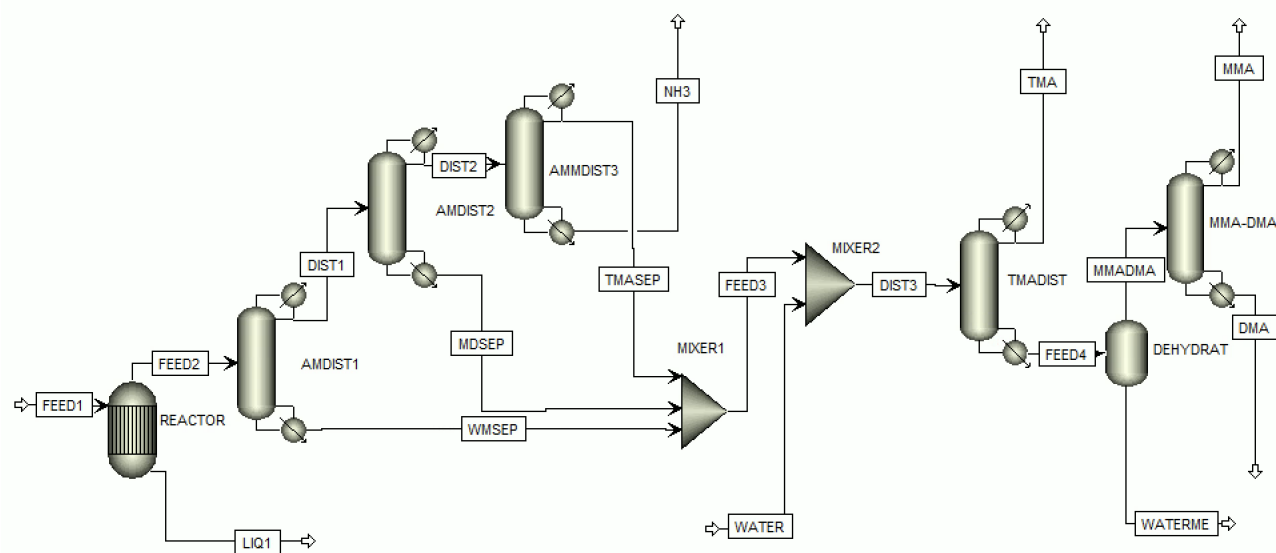


Figure 7: Previous Overall Process Flow Diagram without recycle stream and pumps and heaters, with stream tables in [Tab. 20](#).

Table 20: Previous Process Flow Diagram Stream Tables and Operating Conditions. Temperature in °C, pressure in psia.

Component	Feed1	Liq1	Feed2	Wmsep	Dist1
(tonne/oper-year)					
Ammonia	130 000	76 900.80	0	1393.8	68 296
Methanol	130 000	8219.58	0	4813.2	0
Water	0	68 469.43	0	61 951	0
DMA	0	25 151.20	0	39.0	25 895
MMA	0	77 568.01	0	1848.3	59 761.9
TMA	0	3690.98	0	$3.0 \cdot 10^{-4}$	6001.3
Total	0	0	0	70 045	159 955
Temperature	0	0	0	130	84
Pressure	0	0	0	455	450

(a) First subtable

Component	Mdsep	Dist2	Tmasep	Nh3	Feed3
(tonne/oper-year)					
Ammonia	68.296	68 228	68.228	68 160	1530.3
Methanol	0	0	0	0	4813.2
Water	0	0	0	0	61 951
DMA	25 636	258.95	3.65	255.30	25 679
MMA	58 785	976.687	0.78	975.91	60 634
TMA	$2.3 \cdot 10^{-8}$	6001.3	5941.3	60.01	5941.3
Total	84 490	75 465	6013.9	69 451	160 549
Temperature	85	45	94	46	84
Pressure	255	250	250	255	250

(b) Second subtable

Table 20: Previous Process Flow Diagram Stream Tables and Operating Conditions. Temperature in °C, pressure in psia (continued)

Component	Waterin	Dist3	Feed4	Tma	Waterme	Mmadma
(tonne/oper-year)						
Ammonia	0	1530.3	4.205	0	0.08	0
Methanol	0	4813.2	691.8	0	156.8	534.9
Water	6	61 957	56 271	0	55 197.6	1073.4
DMA	0	25 679	26 290.5	0	2.3	26 288.2
MMA	0	60 634	11 531.1	0	35.9	11 495.1
TMA	0	5941.2	50.5	0	0	50.5
Total	6	160 555	94 839.1	0	55 392.8	39 446.3
Temperature	25	80	20	0	40	40
Pressure	255	255	290	0	14.6	14.6

(c) Third subtable

Component	Dma	Mma
(tonne/oper-year)		
Ammonia	4.1	4.1
Methanol	534.9	0.0018
Water	1073.4	0.001 29
DMA	26 025.3	262.9
MMA	45.98	11 449.2
TMA 24.6	25.9	
Total	27 704.3	11 742
Temperature	67.6	47
Pressure	101.5	101.5

(d) Fourth subtable

Table 21: Caption

10.8 Pressure and Temperature

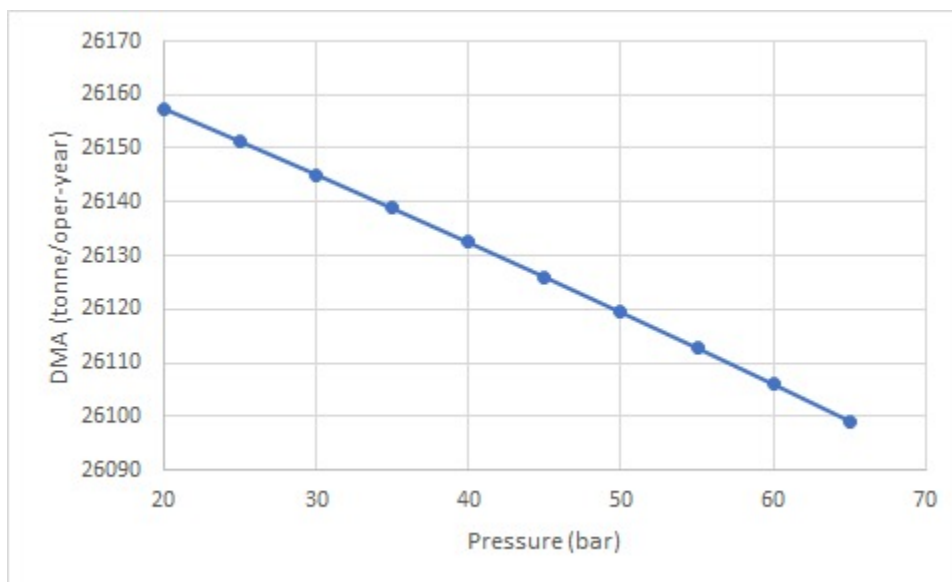


Figure 8: Temperature is kept at 600 K throughout this analysis. A linear downward trend is observed as pressure increase when ratio of $\text{NH}_3/\text{MeOH} = 1$.

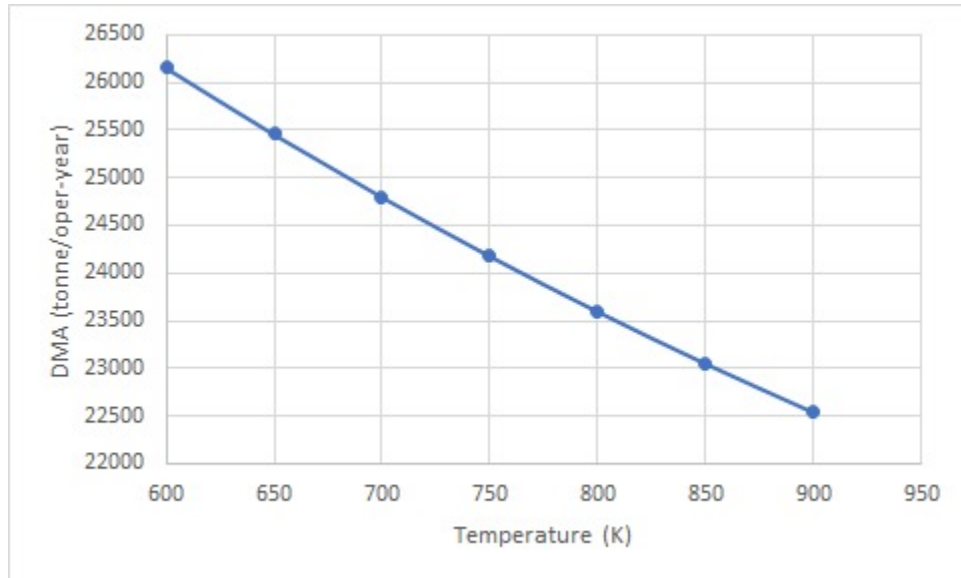


Figure 9: Pressure is constant at 20 bar with ratio of $\text{NH}_3/\text{MeOH} = 1$. A linear with a slight bend downward trend can also be observed as temperature increases. Due to this slight bend in the trend, a power of some number needs to be determined to create a straight linear trend.

Table 22: Overall energy balance for the entire DMA production system

Equipment	Energy Required (Btu/hr)		
Reactor	-8,039,106.05		
Methanol	4,813.2	4,809.06	4.14
Water	61,951.1	61,951.1	7.91e-06
MMA	60,634.3	57,980.8	2,653.53
DMA	25,679	22,984.7	2,694.36
TMA	5,941.25	0.03	5,941.22

10.9 HAZOP Analysis Tables

Line - Feed

Intention - Feed in fresh materials to the system

Guide Word	Deviation	Cause	Consequences and Actions
LESS OF	Flow	Leaks present and/or clogs in feed line	Less feed going into the reactor, loss of product and possible overheating, a barometer or volumetric flow gauge calibrated for the desired feedstock could be implemented
AS WELL AS	Composition	Leaks present or backflow from reactor	Incorrect concentrations of feed going into the reactor and/or contamination, concentration of trace compounds/main feed components could be measured with probes
MORE	Pressure	Backflow from reactor, clogs in feed lines	Over-pressurizing the reactor and/or the lines, leading to possible breakage. Pressure sensor should be installed at the inlet and outlet of the reactor

Vessel - Mixer (B7)

Intention - to mix the Feed and Rec3 lines before they enter the reactor.

MORE	Pressure	Spike in feed rates	Put a max flow rate valve at the inlet of the mixer
LESS	Pressure	Dip in feed rates and/or leakages, also if the impellers are stuck the pressure drop across them will go down	Pressure gauge at the inlet of the reactor to measure/monitor the pressure drop between the mixer and the reactor

Line - FeedRCT

Intention - Feed in fresh materials to the Reactor

LESS OF	Flow	Leaks present and/or clogs in feed line	Less feed going into the reactor, loss of product and possible overheating, a barometer or volumetric flow gauge calibrated for the desired feedstock could be implemented
AS WELL AS	Composition	Leaks present or backflow from reactor	Incorrect concentrations of feed going into the reactor and/or contamination, concentration of trace compounds/main feed components could be measured with probes

Vessel - Reacto

Intention - to react the Feed Streams and recycling streams effectively and efficiently in order to maximize production of our product

MORE	Pressure	Spike in reactor feed rates	Put a max flow rate valve at the inlet of the mixer
LESS	Pressure	Dip in reactor feed rates and/or leakages, also if the impellers are stuck the pressure drop across them will go down	Pressure gauge at the inlet of the reactor to measure/monitor the pressure drop between the mixer and the reactor
MORE	Temperature	The reactor could overheat, altering the rate of formation of the product and posing a safety concern to those working nearby	The product will not be as pure and the equipment will likely wear out faster, and at worst it could explode or melt. Installing Temperature and Pressure sensors would solve this problem and the above one

Line - Vapor

Intention - transports the vapor product from the reactor to the first ammonium separations column

LESS OF	Flow	Leaks present and/or clogs in line	Less feed going into/out of the reactor, loss of product and possible downstream issues a barometer or volumetric flow gauge calibrated for the desired feedstock could be implemented
AS WELL AS	Composition	Leaks present or unexpected reactions occurring in reactor	Incorrect concentrations of feed going into the reactor and/or contamination, concentration of trace compounds/main feed components could be measured with probes

Line - Liquid

Omitted because it simply exists so that Aspen doesn't error, there is nothing actually flowing in that line

Vessel - Amm1

Intention - to provide an initial separation of the products of our reactor in order to remove as much excess ammonium as possible

MORE	Pressure	Spike in column feed rates	Put a max flow rate valve at the inlet of the column
LESS	Pressure	Dip in column feed rates and/or leakages, also if the impellers are stuck the pressure drop across them will go down	Pressure gauge at the inlet of the column to measure/monitor the pressure drop upon entering the column
MORE	Temperature	Faulty temperature reading or broken heater that's always on full blast	The product will not be as pure and the equipment will likely wear out faster, and at worst it could explode or melt or dry out the column. Installing Temperature and Pressure sensors would solve this problem and the above one
LESS	Temperature	Loss of power/faulty heater/reboiler	Could flood the column, and cause incorrect separations skewed towards the bottoms streams. Installing temperature sensors, a liquid volume height measurer at the bottom of the column and an emergency valve that opens to a secondary holding container if the height alarm is triggered
LESS	Flow	Leak in the inlet line	Possible column drying up, skewed separations ratios between exit streams, installing a flowmeter on the inlet stream would mitigate this
MORE	Flow	Unexpected spike in feed rate	Could also flood the column and cause incorrect separations ratios. Installing a flowmeter with a valve that allows operations technicians to manually lower the flow rate to acceptable levels would be advised

Line - Amm1Dist

Intention - transports the vapor product from the first ammonia column to the second one

LESS OF	Flow	Leaks present and/or clogs in line	Less feed going out of the first ammonia column, loss of product and possible downstream issues. a barometer or volumetric flow gauge calibrated for the desired feedstock could be implemented
AS WELL AS	Composition	Leaks present or unexpected reactions occurring in reactor	Incorrect exit stream concentrations from the first ammonia column for whatever reason and/or contamination, concentration of trace compounds/main feed components could be measured with probes, as well as other measurements such as conductivity or density or viscosity to sense deviations

Line - Amm1Bot

Intention - transports the bottoms product of the first ammonia column to be further processed to remove TMA and separate out MMA and DMA

LESS OF	Flow	Leaks present and/or clogs in line	Less feed going out of the first ammonia column, loss of product and possible downstream issues. a barometer or volumetric flow gauge calibrated for the desired feedstock could be implemented
AS WELL AS	Composition	Leaks present or unexpected reactions occurring in column	Incorrect exit stream concentrations from the first ammonia column for whatever reason and/or contamination, concentration of trace compounds/main feed components could be measured with probes, as well as other measurements such as conductivity or density or viscosity to sense deviations

Vessel - Amm2

Intention - to further remove ammonium from our product streams

MORE	Pressure	Spike in column feed rates	Put a max flow rate valve at the inlet of the column
LESS	Pressure	Dip in column feed rates and/or leakages, also if the impellers are stuck the pressure drop across them will go down	Pressure gauge at the inlet of the column to measure/monitor the pressure drop upon entering the column
MORE	Temperature	Faulty temperature reading or broken heater that's always on full blast	The product will not be as pure and the equipment will likely wear out faster, and at worst it could explode or melt or dry out the column. Installing Temperature and Pressure sensors would solve this problem and the above one

LESS	Temperature	Loss of power/faulty heater/reboiler	Could flood the column, and cause incorrect separations skewed towards the bottoms streams. Installing temperature sensors, a liquid volume height measurer at the bottom of the column and an emergency valve that opens to a secondary holding container if the height alarm is triggered
LESS	Flow	Leak in the inlet line	Possible column drying up, skewed separations ratios between exit streams, installing a flowmeter on the inlet stream would mitigate this
MORE	Flow	Unexpected spike in feed rate	Could also flood the column and cause incorrect separations ratios. Installing a flowmeter with a valve that allows operations technicians to manually lower the flow rate to acceptable levels would be advised

Line - **Amm2Dist**

Intention - transports the vapor product from the second ammonia column to the third one

LESS OF	Flow	Leaks present and/or clogs in line	Less outlet going out of the first ammonia column, loss of product and possible downstream issues. a barometer or volumetric flow gauge calibrated for the desired feedstock could be implemented
AS WELL AS	Composition	Leaks present or unexpected reactions occurring in reactor	Incorrect exit stream concentrations from the second ammonia column for whatever reason and/or contamination, concentration of trace compounds/main feed components could be measured with probes, as well as other measurements such as conductivity or density or viscosity to sense deviations

Line - **Amm2Bot**

Intention - transports the bottoms product of the second ammonia column to be further processed to remove TMA and separate out MMA and DMA

LESS OF	Flow	Leaks present and/or clogs in line	Less feed going out of the second ammonia column, loss of product and possible downstream issues. a barometer or volumetric flow gauge calibrated for the desired feedstock could be implemented
AS WELL AS	Composition	Leaks present or unexpected reactions occurring in column	Incorrect exit stream concentrations from the second ammonia column for whatever reason and/or contamination, concentration of trace compounds/main feed components could be measured with probes, as well as other measurements such as conductivity or density or viscosity to sense deviations

Vessel - **Amm3**

Intention - to further remove ammonium from our product streams

MORE	Pressure	Spike in column feed rates	Put a max flow rate valve at the inlet of the column
LESS	Pressure	Dip in column feed rates and/or leakages, also if the impellers are stuck the pressure drop across them will go down	Pressure gauge at the inlet of the column to measure/monitor the pressure drop upon entering the column
MORE	Temperature	Faulty temperature reading or broken heater that's always on full blast	The product will not be as pure and the equipment will likely wear out faster, and at worst it could explode or melt or dry out the column. Installing Temperature and Pressure sensors would solve this problem and the above one
LESS	Temperature	Loss of power/faulty heater/reboiler	Could flood the column, and cause incorrect separations skewed towards the bottoms streams. Installing temperature sensors, a liquid volume height measurer at the bottom of the column and an emergency valve that opens to a secondary holding container if the height alarm is triggered
LESS	Flow	Leak in the inlet line	Possible column drying up, skewed separations ratios between exit streams, installing a flowmeter on the inlet stream would mitigate this
MORE	Flow	Unexpected spike in feed rate	Could also flood the column and cause incorrect separations ratios. Installing a flowmeter with a valve that allows operations technicians to

			manually lower the flow rate to acceptable levels would be advised
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Line - Amm3Dist Intention - transports the vapor product from the third ammonia column to be further separated into our desired products			
LESS OF	Flow	Leaks present and/or clogs in line	Less outlet going out of the third ammonia column, loss of product and possible downstream issues. a barometer or volumetric flow gauge calibrated for the desired feedstock could be implemented
AS WELL AS	Composition	Leaks present or unexpected reactions occurring in reactor	Incorrect exit stream concentrations from the third ammonia column for whatever reason and/or contamination, concentration of trace compounds/main feed components could be measured with probes, as well as other measurements such as conductivity or density or viscosity to sense deviations

Line - Amm3Bot Intention - transports the bottoms product of the third ammonia column to be recycled into feed for the reactor			
LESS OF	Flow	Leaks present and/or clogs in line	Less feed going out of the third ammonia column, loss of product and possible downstream issues. a barometer or volumetric flow gauge calibrated for the desired feedstock could be implemented
AS WELL AS	Composition	Leaks present or unexpected reactions occurring in column	Incorrect exit stream concentrations from the third ammonia column for whatever reason and/or contamination, concentration of trace compounds/main feed components could be measured with probes, as well as other measurements such as conductivity or density or viscosity to sense deviations

Vessel - Mixer Intention - to combine the product heavy streams Amm3Dist, Amm2Bot, and Amm1Bot in order to be collectively pumped up to a desired pressure before being separated into the three different methylamines			
LESS	Flow	Leaks or blockages in inlet lines	Possible deadheading of the pump, increased wear and decreased efficiency, if severe enough possible drying out. Flowmeters/pressure gauges and valves at the inlet of the mixer are recommended
LESS	Pressure	Improper upstream separations, leaks or blockages in inlet lines	Possible deadheading of the pump, increased wear and decreased efficiency, Flowmeters/pressure gauges and valves at the inlet of the mixer are recommended
MORE	Pressure	Improper upstream separations	Unless the inlet pressure is very close to or more than the pump outlet pressure, this isn't too much of an issue, however too high of pressure can warp the mixer

Line - ToPump Intention - transports the product heavy methylamines mixture from the mixer to a pump to be pressurized before further separation			
LESS OF	Flow	Leaks present and/or clogs in line	Less feed going out of the third ammonia column, loss of product and possible downstream issues. a barometer or volumetric flow gauge calibrated for the desired feedstock could be implemented
AS WELL AS	Composition	Leaks present or unexpected reactions occurring in column	Incorrect exit stream concentrations/pressure from the mixer for whatever reason and/or contamination, concentration of trace compounds/main feed components could be measured with probes, as well as other measurements such as conductivity or density or viscosity to sense deviations
NO	Pressure	Leaks or clogs in the inlet lines	At low enough pressures, the liquid feed will start to evaporate, leading to deadheading of the pump which is highly damaging. Automatic shutoff valves connected to flowmeters/in stream pressure sensors are recommended

Vessel - Pump1 Intention - to increase the pressure on the inlet stream to the TMA column in order to prevent drying up and flash calculation errors			
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LESS	Pressure	Leaks, clogs, upstream conditions requiring lowered feed rates	Could cause undue strain and wear, and possible deadheading Emergency shutoff valves/switches recommended
NO	Pressure	Leaks, clogs, upstream conditions requiring halted feed rates	Will cause extreme wear and strain, and will deadhead the pump. Emergency shutoff valves/switches recommended
MORE	Temperature	Heat from friction in the mixer/lines	Could cause some of the inlet to evaporate, , which will make the pump have to work extremely hard to function. Emergency shutoff valves/switches recommended

Line - **ToTMACol**

Intention - transports the pressurized product stream to the TMA column

LESS OF	Flow	Leaks present and/or clogs in line	Less feed going out of the third ammonia column, loss of product and possible downstream issues. a barometer or volumetric flow gauge calibrated for the desired feedstock could be implemented
AS WELL AS	Composition	Leaks present or unexpected reactions occurring in column	Incorrect exit stream concentrations/pressure from the mixer for whatever reason and/or contamination, concentration of trace compounds/main feed components could be measured with probes, as well as other measurements such as conductivity or density or viscosity to sense deviations

Vessel - **TMA Column (B4)**

Intention - to extract TMA from the azeotropic methylamines mixture

MORE	Pressure	Spike in column feed rates, clogs in exit lines, overheating	Put a max flow rate valve at the inlet of the column, have an automatic shutoff for the reboiler and/or an emergency pressure release valve if necessary
LESS	Pressure	Dip in column feed rates and/or leakages, also if the heater gets turned down too low and there isn't enough vapor in the column	Probably drying out of the column, skewed exit stream concentrations. pressure gauge at the inlet of the column to measure/monitor the pressure drop upon entering the column
MORE	Temperature	Faulty temperature reading or broken heater that's always on full blast	The product will not be as pure and the equipment will likely wear out faster, and at worst it could explode or melt or dry out the column. Installing Temperature and Pressure sensors would solve this problem and the above one
LESS	Temperature	Loss of power/faulty heater/reboiler	Could flood the column, and cause incorrect separations skewed towards the bottoms streams. Installing temperature sensors, a liquid volume height measurer at the bottom of the column and an emergency valve that opens to a secondary holding container if the height alarm is triggered
LESS	Flow	Leak in the inlet line	Possible column drying up, skewed separations ratios between exit streams, installing a flowmeter on the inlet stream would mitigate this
MORE	Flow	Unexpected spike in feed rate	Could also flood the column and cause incorrect separations ratios. Installing a flowmeter with a valve that allows operations technicians to manually lower the flow rate to acceptable levels would be advised

Line - **TMATop**

Intention - transports the TMA heavy stream to be further separated

LESS OF	Flow	Leaks present and/or clogs in line	Less feed going out of the TMA column, loss of product and possible downstream issues. a barometer or volumetric flow gauge calibrated for the desired feedstock could be implemented
AS WELL AS	Composition	Leaks present or unexpected reactions occurring in column	Incorrect exit stream concentrations/pressure from the column outlets for whatever reason and/or contamination, concentration of trace compounds/main feed components could be measured with probes, as well as other measurements such as conductivity or density or viscosity to sense

			deviations
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Line - **Bottoms**

Intention - transports the TMA free stream to be further separated

LESS OF	Flow	Leaks present and/or clogs in line	Less feed going out of the TMA column, loss of product and possible downstream issues. a barometer or volumetric flow gauge calibrated for the expected feedstock could be implemented
AS WELL AS	Composition	Leaks present or unexpected reactions occurring in column	Incorrect exit stream concentrations/pressure from the column outlets for whatever reason and/or contamination, concentration of trace compounds/main feed components could be measured with probes, as well as other measurements such as conductivity or density or viscosity to sense deviations

Vessel - **Hydrat**

Intention - to extract excess water and methanol from the product streams so later downstream MMA and DMA can be separated

MORE	Pressure	Spike in column feed rates, clogs in exit lines, overheating	Put a max flow rate valve at the inlet of the column, have an automatic shutoff for the reboiler and/or an emergency pressure release valve if necessary
LESS	Pressure	Dip in column feed rates and/or leakages, also if the heater gets turned down too low and there isn't enough vapor in the column	Probably drying out of the column, skewed exit stream concentrations. pressure gauge at the inlet of the column to measure/monitor the pressure drop upon entering the column
MORE	Temperature	Faulty temperature reading or broken heater that's always on full blast	The product will not be as pure and the equipment will likely wear out faster, and at worst it could explode or melt or dry out the column. Installing Temperature and Pressure sensors would solve this problem and the above one
LESS	Temperature	Loss of power/faulty heater/reboiler	Could flood the column, and cause incorrect separations skewed towards the bottoms streams. Installing temperature sensors, a liquid volume height measurer at the bottom of the column and an emergency valve that opens to a secondary holding container if the height alarm is triggered
LESS	Flow	Leak in the inlet line	Possible column drying up, skewed separations ratios between exit streams, installing a flowmeter on the inlet stream would mitigate this
MORE	Flow	Unexpected spike in feed rate	Could also flood the column and cause incorrect separations ratios. Installing a flowmeter with a valve that allows operations technicians to manually lower the flow rate to acceptable levels would be advised

Line - **MMADMA**

Intention - transports the MMA/DMA Distillate stream from Hydrat to be further separated

LESS OF	Flow	Leaks present and/or clogs in line	Less feed going out of the Hydrat flash, loss of product and possible downstream issues. a barometer or volumetric flow gauge calibrated for the desired feedstock could be implemented
AS WELL AS	Composition	Leaks present or unexpected reactions occurring in column	Incorrect exit stream concentrations/pressure from the column outlets for whatever reason and/or contamination, concentration of trace compounds/main feed components could be measured with probes, as well as other measurements such as conductivity or density or viscosity to sense deviations

Line - **Water-ME**

Intention - transports the excess water and Methanol condensate from the flash drum

LESS OF	Flow	Leaks present and/or clogs in line	Less feed going out of the Hydrat flash drum, loss of product and possible downstream issues. a barometer or volumetric flow gauge calibrated for the expected feedstock could be implemented
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AS WELL AS	Composition	Leaks present or unexpected reactions occurring in column	Incorrect exit stream concentrations/pressure from the drum outlets for whatever reason and/or contamination, concentration of trace compounds/main feed components could be measured with probes, as well as other measurements such as conductivity or density or viscosity to sense deviations
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Vessel - **MMA-DMA**

Intention - to separate MMA and DMA from each other

MORE	Pressure	Spike in column feed rates, clogs in exit lines, overheating	Put a max flow rate valve at the inlet of the column, have an automatic shutoff for the reboiler and/or an emergency pressure release valve if necessary
LESS	Pressure	Dip in column feed rates and/or leakages, also if the heater gets turned down too low and there isn't enough vapor in the column	Probably drying out of the column, skewed exit stream concentrations. pressure gauge at the inlet of the column to measure/monitor the pressure drop upon entering the column
MORE	Temperature	Faulty temperature reading or broken heater that's always on full blast	The product will not be as pure and the equipment will likely wear out faster, and at worst it could explode or melt or dry out the column. Installing Temperature and Pressure sensors would solve this problem and the above one
LESS	Temperature	Loss of power/faulty heater/reboiler	Could flood the column, and cause incorrect separations skewed towards the bottoms streams. Installing temperature sensors, a liquid volume height measurer at the bottom of the column and an emergency valve that opens to a secondary holding container if the height alarm is triggered
LESS	Flow	Leak in the inlet line	Possible column drying up, skewed separations ratios between exit streams, installing a flowmeter on the inlet stream would mitigate this
MORE	Flow	Unexpected spike in feed rate	Could also flood the column and cause incorrect separations ratios. Installing a flowmeter with a valve that allows operations technicians to manually lower the flow rate to acceptable levels would be advised

Line - **MMIAINT**

Intention - transports the MMAheave distillate for one final separations step to maximise our recycle streams

LESS OF	Flow	Leaks present and/or clogs in line	Less feed going out of the distillation column, loss of product and possible downstream issues. a barometer or volumetric flow gauge calibrated for the desired feedstock could be implemented
AS WELL AS	Composition	Leaks present or unexpected reactions occurring in column	Incorrect exit stream concentrations/pressure from the column outlets for whatever reason and/or contamination, concentration of trace compounds/main feed components could be measured with probes, as well as other measurements such as conductivity or density or viscosity to sense deviations

Line - **DMA**

Intention - transports the desired Product to where it can be packaged/otherwise processed

LESS OF	Flow	Leaks present and/or clogs in line	Less feed going out of the MMA-DMA distillation column, loss of product and possible downstream issues. a barometer or volumetric flow gauge calibrated for the expected feedstock could be implemented
AS WELL AS	Composition	Leaks present or unexpected reactions occurring in column	Incorrect exit stream concentrations/pressure from the column outlets for whatever reason and/or contamination, concentration of trace compounds/main feed components could be measured with probes, as well as other measurements such as conductivity or density or viscosity to sense deviations

Vessel - **B1**

Intention - to boil off undesired impurities from out MMA product

MORE	Pressure	Spike in drum feed rates, clogs in exit lines, overheating	Put a max flow rate valve at the inlet of the drum, have an automatic shutoff for the reboiler and/or an emergency pressure release valve if necessary
LESS	Pressure	Dip in drum feed rates and/or leakages, also if the heater gets turned down too low and there isn't enough vapor in the column	Probably drying out of the drum, skewed exit stream concentrations. pressure gauge at the inlet of the column to measure/monitor the pressure drop upon entering the column
MORE	Temperature	Faulty temperature reading or broken heater that's always on full blast	The product will not be as pure and the equipment will likely wear out faster, and at worst it could explode or melt or dry out the drum. Installing Temperature and Pressure sensors would solve this problem and the above one
LESS	Temperature	Loss of power/faulty heater/reboiler	Could flood the drum, and cause incorrect separations skewed towards the bottoms streams. Installing temperature sensors, a liquid volume height measurer at the bottom of the column and an emergency valve that opens to a secondary holding container if the height alarm is triggered
LESS	Flow	Leak in the inlet line	Possible drum drying up, skewed separations ratios between exit streams, installing a flowmeter on the inlet stream would mitigate this
MORE	Flow	Unexpected spike in feed rate	Could also flood the drum and cause incorrect separations ratios. Installing a flowmeter with a valve that allows operations technicians to manually lower the flow rate to acceptable levels would be advised

Line - MMA

Intention - transports the other Product to where it can be packaged/otherwise processed

LESS OF	Flow	Leaks present and/or clogs in line	Less feed going out of the MMA flash drum distillation column, loss of product and possible downstream issues. a barometer or volumetric flow gauge calibrated for the expected feedstock could be implemented
AS WELL AS	Composition	Leaks present or unexpected reactions occurring in column	Incorrect exit stream concentrations/pressure from the drum outlets for whatever reason and/or contamination, concentration of trace compounds/main feed components could be measured with probes, as well as other measurements such as conductivity or density or viscosity to sense deviations

Vessel -B2

Intention - to further remove TMA so it can be sold as a byproduct

MORE	Pressure	Spike in column feed rates, clogs in exit lines, overheating	Put a max flow rate valve at the inlet of the column, have an automatic shutoff for the reboiler and/or an emergency pressure release valve if necessary
LESS	Pressure	Dip in column feed rates and/or leakages, also if the heater gets turned down too low and there isn't enough vapor in the column	Probably drying out of the column, skewed exit stream concentrations. pressure gauge at the inlet of the column to measure/monitor the pressure drop upon entering the column
MORE	Temperature	Faulty temperature reading or broken heater that's always on full blast	The product will not be as pure and the equipment will likely wear out faster, and at worst it could explode or melt or dry out the column. Installing Temperature and Pressure sensors would solve this problem and the above one
LESS	Temperature	Loss of power/faulty heater/reboiler	Could flood the column, and cause incorrect separations skewed towards the bottoms streams. Installing temperature sensors, a liquid volume height measurer at the bottom of the column and an emergency valve that opens to a secondary holding container if the height alarm is triggered
LESS	Flow	Leak in the inlet line	Possible column drying up, skewed separations ratios between exit streams, installing a flowmeter on the inlet stream would mitigate this

MORE	Flow	Unexpected spike in feed rate	Could also flood the column and cause incorrect separations ratios. Installing a flowmeter with a valve that allows operations technicians to manually lower the flow rate to acceptable levels would be advised
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Line - TMA

Intention - transports the TMA byproduct to where it can be packaged/otherwise processed

LESS OF	Flow	Leaks present and/or clogs in line	Less feed going out of the B2 distillation column, loss of product and possible downstream issues. a barometer or volumetric flow gauge calibrated for the expected feedstock could be implemented
AS WELL AS	Composition	Leaks present or unexpected reactions occurring in column	Incorrect exit stream concentrations/pressure from the column outlets for whatever reason and/or contamination, concentration of trace compounds/main feed components could be measured with probes, as well as other measurements such as conductivity or density or viscosity to sense deviations

Line - RECTMA

Intention - transports rectified distillate from B2 to be recycled back into the reactor

LESS OF	Flow	Leaks present and/or clogs in line	Less feed going out of the distillation column, loss of product and possible downstream issues. a barometer or volumetric flow gauge calibrated for the desired feedstock could be implemented
AS WELL AS	Composition	Leaks present or unexpected reactions occurring in column	Incorrect exit stream concentrations/pressure from the column outlets for whatever reason and/or contamination, concentration of trace compounds/main feed components could be measured with probes, as well as other measurements such as conductivity or density or viscosity to sense deviations

Line - REC1

Intention - transports the recycle stream from the MMA drum to be mixed with RECTMA

LESS OF	Flow	Leaks present and/or clogs in line	Less feed going out of the MMA drum, loss of product and possible downstream issues. a barometer or volumetric flow gauge calibrated for the desired feedstock could be implemented
AS WELL AS	Composition	Leaks present or unexpected reactions occurring in column	Incorrect exit stream concentrations/pressure from the drum outlets for whatever reason and/or contamination, concentration of trace compounds/main feed components could be measured with probes, as well as other measurements such as conductivity or density or viscosity to sense deviations

Line - S2

Intention - transports the mixed MMA and TMA column excesses to be mixed and pre-heated at B3

LESS OF	Flow	Leaks present and/or clogs in line	Less feed going out of the MMA-TMA mixer, loss of product and possible downstream issues. a barometer or volumetric flow gauge calibrated for the desired feedstock could be implemented
AS WELL AS	Composition	Leaks present or unexpected reactions occurring in column	Incorrect exit stream concentrations/pressure from the mixer outlets for whatever reason and/or contamination, concentration of trace compounds/main feed components could be measured with probes, as well as other measurements such as conductivity or density or viscosity to sense deviations

Vessel - Mixer (B3)

Intention - to mix the Ammonium column excesses with the downstream recycling streams and pre-heated before it enters the reactor

MORE	Pressure	Spike in feed rates	Put a max flow rate valve at the inlet of the mixer
LESS	Pressure	Dip in feed rates and/or leakages, also if the impellers are stuck the pressure drop across them will go down	Pressure gauge at the inlet of the reactor to measure/monitor the pressure drop between the mixer and the reactor

Line - **REC3**

Intention - transports the finished recycle stream to the B7 mixer to enter the reactor

LESS OF	Flow	Leaks present and/or clogs in line	Less feed going out of the recycle stream mixer, loss of product and possible downstream issues. a barometer or volumetric flow gauge calibrated for the desired feedstock could be implemented
AS WELL AS	Composition	Leaks present or unexpected reactions occurring in column	Incorrect exit stream concentrations/pressure from the mixer outlets for whatever reason and/or contamination, concentration of trace compounds/main feed components could be measured with probes, as well as other measurements such as conductivity or density or viscosity to sense deviations

10.10 Contributions

Table 23: Summary of Contributions per Group Member

	Column/Unit Operation	Other Contributions
Jovan Yaputra	MMA Reactor (Part 1)	Setting up the initial document, Optimizing the reactor and adding recycle streams, Equipment Sizing & Equipment Materials, and setting the Gross margin table
Brian Chu	Ammonia Columns (Part 2)	PFD and PID Controls Diagram, Formatting PFD stream tables, NPV Analysis
Anna Dougherty	TMA Column (Part 3)	HAZOP Analysis, Stream mass flow and Energy tables, and sundry explanations/formatting (including Formatting the document, writing the Introduction, Abstract, Basis of Design, and Project Plan Sections)
Kunpeng Yu	Dehydration and MMA/DMA Separation Columns (Part 4)	Economic Analysis and Environmental Considerations